



Statistical Sampling and Data Collection

Worksheet 1

Statistics Chapter 1: Statistical Sampling and Data Collection. Answer all questions. Show your working clearly. Total: 58 marks. Time: 70 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 1. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- When you criticise a sampling method, say **which group** is over- or under-represented and **why**.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Populations, samples and sampling frames	17	
B Sampling techniques	20	
C Types of data and the large data set	17	
Roar-level challenge	4	
Total	58	

Section A Populations, samples and sampling frames

17 marks

1. Explain what is meant by (a) a *population*, (b) a *sampling frame*. [2]

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2. A factory makes batteries and wants to find out how long they last before running flat. [2]
Give two reasons why the factory should use a sample rather than a census.

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3. A town council wants to find out the views of the town's adult residents on a proposed new park. [3]

- (a) Suggest a suitable sampling frame.
- (b) Give one reason why your sampling frame might not be perfect.
- (c) Explain why the council is likely to use a sample rather than a census.

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4. A headteacher wants to know what the 1200 students at her school think of the school meals. She asks 50 students who are waiting in the lunch queue. [3]
Explain why this sample is likely to be biased, stating the direction in which the results may be biased.

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5. A gym has 2000 members. The manager wants to survey members about how satisfied they are with the gym. He uses a list of the members who visited the gym in the last week as his sampling frame.

(a) State the population in this survey. [1]

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(b) Explain why this sampling frame could lead to biased results. [2]

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(c) Suggest a better sampling frame, and give a reason why it is better. [2]

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(d) Give one advantage and one disadvantage of the manager carrying out a census of all members instead. [2]

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(Total for Question 5 is 7 marks)



Section B Sampling techniques

20 marks

6. Describe how to take a simple random sample of 30 students from a school of 400 students. [2]

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7. A systematic sample of 20 is to be taken from a list of 500 customers. [3]

- (a) Describe how the sample should be taken.
- (b) Give one disadvantage of systematic sampling.

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8. A sixth form has 240 boys and 160 girls. A stratified sample of 50 students is taken, stratified by gender. [3]
Find the number of boys and the number of girls in the sample.

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9. A company has three departments with 120, 85 and 45 employees. A stratified sample of 40 employees is to be taken. [4]

(a) Find the number of employees that should be sampled from each department.

(b) Describe how the employees in the largest department should then be chosen.

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10. A sports club has 84 junior members, 156 adult members and 60 senior members.

(a) Explain the difference between quota sampling and stratified sampling. [2]

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(b) Give one advantage and one disadvantage of using quota sampling for this club. [2]

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(c) A stratified sample of 25 members is taken. Find how many juniors, adults and seniors should be in the sample. [3]

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(d) Explain why an opportunity sample of members in the club café would not be suitable. [1]

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(Total for Question 10 is 8 marks)

**Section C Types of data and the large data set**

17 marks

11. State whether each variable is qualitative or quantitative. If quantitative, state whether it is discrete or continuous. [2]

(a) The number of goals scored in a football match.

(b) The eye colour of a student.

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12. For each of the following weather variables, state whether it is discrete or continuous. [3]

(a) Daily maximum temperature ($^{\circ}\text{C}$).

(b) Daily total sunshine (hours).

(c) The number of days with rain in a month.

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13. Amira collects data on travel times by asking her classmates. Ben uses travel data published by the government. State, with a reason, whether each student is using primary or secondary data. [2]

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14. Explain why shoe size is a discrete variable even though it can take values such as $7\frac{1}{2}$, whereas a 100 m sprint time recorded to the nearest 0.1 seconds is a continuous variable. [3]

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15. The table shows an extract from the large data set for Leuchars in May 1987.

Date	Daily total rainfall (mm)	Daily mean windspeed (kn)	Cloud cover (oktas)
1/5/1987	0.0	8	6
2/5/1987	tr	11	7
3/5/1987	4.6	n/a	8
4/5/1987	1.2	14	5

- (a) State whether *cloud cover* and *daily total rainfall* are discrete or continuous, giving a reason for each. [2]

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- (b) In the large data set, “tr” means a trace of rain (less than 0.05 mm). Explain how you would treat this value when calculating the mean daily rainfall, and why. [2]

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- (c) A student uses the May 1987 data to make conclusions about rainfall at Leuchars in all years. Give two reasons why this might not be reliable. [2]

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- (d) Explain how the value “n/a” should be treated when calculating the mean windspeed. [1]

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(Total for Question 15 is 7 marks)



ROAR-LEVEL CHALLENGE

4 marks

16. A school has 1500 students in five year groups, with 320, 310, 300, 290 and 280 students. A stratified sample of 60 students is taken, stratified by year group. Find the number of students sampled from each year group, making sure your numbers add up to 60, and explain why a stratified sample is better than a simple random sample here. [4]

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END OF WORKSHEET

Total: 58 marks